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**Rapidity
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**Electroweak
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**THE HQIV PROOF SPINE: MACHINE-CHECKED
SYNTHESIS OF FLAVOR, ELECTROWEAK,
STRONG-SECTOR, AND COSMOLOGICAL READOUTS
FROM THE DISCRETE OCTONION CARRIER**

STEVEN ETTINGER JR

ABSTRACT. We present a programme-level synthesis of the HQIV physics stack: a self-reinforcing, machine-checked proof ladder anchored on the discrete null lattice and octonionic carrier at `referenceM=4`, with horizon coefficient $\gamma = 2/5$ and curvature imprint $\alpha = 3/5$. The capstone certificate `proofSpineInventory_holds (Hqiv.Physics.ProofSpineInventory)` bundles seven phases—Fano holonomy overlap, CKM holonomy, PMNS holonomy, rare-decay and differential observables, electroweak precision and discrete SMEFT export, collider PDF/unitarity witnesses, and cosmological perturbation records—plus a Sievers glueball bridge on the strong sector. This document positions the targeted Tier papers (flavor, HEP decay, TUFT+SM, glueball curvature, baryogenesis) relative to the verified spine, highlights concrete phenomenology (Jarlskog correction ratio, longitudinal shifts on P'_5 and A_{FB}), and lists honest open items (PMNS closure, form-factor-level longitudinal insertion, formal QFT gaps). Comparison PDG and facility numerals remain outside Lean proof hypotheses throughout.

1. INTRODUCTION: TWO AXIOMS, ONE CARRIER

HQIV organizes physics around (i) a discrete light-cone information budget and (ii) informational monogamy on the octonion carrier. Masses, mixings, and decay readouts are *discharges* of holonomy and Casimir structure on Hopf shells—not fitted post-hoc parameters. The proof spine map (`docs/PROOF_SPINE_MAP.md`) records module status (*proved*, *witness*, *scaffold*); this paper is the narrative layer for a wider audience.

2. ARCHITECTURE OF THE PROOF LADDER

Layer 0: OctonionicLightCone, HQVMetric
Layer 1: FanoHolonomyOverlap, FanoMixingMatrix, ReadoutGaugeSeed
Layer 2: CkmHolonomyReadout, PMNSHolonomyReadout, RareDecayReadout, FlavorCPObservable
Layer 3: HiggsSelfCouplingReadout, SMEFTDiscreteExporter
Layer 4: PartonCarrierEvolution, PatchScatteringUnitarity
Layer 5: CosmologicalPerturbationReadout (+ QFT scaffold)
Capstone: proofSpineInventory_holds

Build the flavor closure with `lake build paper_flavor_mixing` (3383 jobs at time of writing).

3. TARGETED PAPERS AND THEIR SPINE HOOKS

Table 1 maps domain-specific manuscripts to Lean modules. The flavour Tier-3 paper is now sufficiently concrete to cite as the CP/mixing anchor; this synthesis does not duplicate its proofs.

3.1. Flavor Tier-3 (machine-checked highlights). The flavour readout paper proves:

- CKM magnitude ledger with row/column unitarity and $\delta_{CP} = 3\pi/32$;
- ledger Jarlskog $J_{\text{ledger}} = 3/80$ from CP-odd holonomy skew;

TABLE 1. HQIV targeted papers mapped to proof-spine phases.

Domain	Tier	Lean spine	Paper fo
Flavor / CP / SMEFT	Tier-3	FlavorDifferentialReadout, ProofSpineInventory	flavor_m
HEP decay multichannel	Tier-2	HepDecayReadout	hep_deca
TUFT + SM Lagrangian	Tier-1	TuftShellChart, HopfShellBeltramiMassBridge	tuft_sm_
Gluon curvature / glueball	Strong	GluonCurvatureArtifact, GluonSieversHopfBridge	gluon_cu
Baryogenesis lock-in	Cosmo	BaryogenesisWitness	baryogere
This synthesis	Overview	proofSpineInventory_holds	proof_sp

TABLE 2. Illustrative phenomenology from the proved slots (comparison numerals outside Lean hypotheses).

Quantity	Interpretation	Value
J_{ledger}	CP-odd holonomy skew (proved)	0.03750
J_{param}	PDG $\sin \delta \sqrt{ V_{us} ^2 V_{cb} ^4} \prod \cos \theta$	$7.81e - 04$
$J_{\text{param}}/J_{\text{ledger}}$	Proved positive correction ratio	0.0208 (2.1%)
$\Delta\phi_{\text{imprint}}$	$\alpha \log(\phi + 1) \Delta\Omega$ at $m = 4$	0.2080
P'_5 shift	Longitudinal factor $1 + \gamma \Delta\phi $	83211ppm
A_{FB} shift	Same factor on angular skeleton	83069ppm

- PDG Jarlskog factorization `ckmJarlskogFromAngles_eq_sin_delta_sin_cos_envelope` with proved positive correction ratio;
- differential skeleton: $P'_5 = 11/400$, $A_{\text{FB}}(B \rightarrow K^* \mu \mu) = 8800/160121$;
- longitudinal amplitude bridge: multiplicative $1 + \gamma|\Delta\phi_{\text{imprint}}|$ correction;
- Sievers bridge: $\kappa_7/\kappa_1 = |V_{cb}|^2/|V_{us}|^2 = 1/4$.

4. PHENOMENOLOGY AT THE REMAINING EDGES

Several quantities are proved as structural identities; their *magnitudes* carry phenomenological content when compared to experiment (outside proof hypotheses). Table 2 summarizes the current illustrative numerics.

Jarlskog correction ratio. The ledger invariant $J_{\text{ledger}} = 3/80$ is the canonical HQIV CP-odd holonomy slot. The standard parameterization product J_{param} evaluates to $\sim 2\%$ of the ledger value at lock-in—they are *not* interchangeable. The proved correction ratio $J_{\text{param}}/J_{\text{ledger}} \approx 0.021$ is therefore a sharp prediction: either experimental comparison must use the ledger skew as the primary HQIV object, or the sin/cos envelope must be promoted to a measured multiplicative factor in facility-facing pipelines.

Longitudinal shift on angular observables. At $m = \text{referenceM} = 4$, the imprint increment $\Delta\phi = \alpha \log(\phi + 1) (\Omega_{m+1} - \Omega_m) \approx 0.21$ yields a longitudinal correction $1 + \gamma|\Delta\phi| \approx 1.08$. Applied to the proved angular skeleton, this moves P'_5 and A_{FB} by $\sim 8 \times 10^4$ ppm ($\sim 8\%$)—large enough to matter at

LHCb precision if the phase-gradient coupling enters the form-factor layer (currently open).

PMNS sector (honest status). PMNS angles and $\delta_{\text{CP}}^{\text{PMNS}} = \pi/5$ are discharged via the T10 overlap wrapper with proved normal ordering; full PMNS Jarlskog closure and sub-percent angle comparison are *lighter* than the CKM side and flagged for the next spine increment.

5. STRONG SECTOR CONVERGENCE

Sievers' octonionic Hopf glueball picture [1] and HQIV's `GluonSieversHopfBridge` share the two-channel drag/stiffness narrative without identifying lattice glueball masses as proof inputs. The consolidation programme treats this as convergent geometric support, not a merged model.

6. OPEN SPINE ITEMS (ROADMAP)

- (1) Promote PMNS from T10 wrapper to full mixing-matrix certificates matching CKM depth.
- (2) Insert longitudinal O-Maxwell coupling into rare-decay form factors (beyond multiplicative scaffold).
- (3) Resolve ledger vs parameterization Jarlskog normalization for facility pipelines.
- (4) Close `ContinuumManyBodyQFTScaffold` renormalization/microcausality gaps.
- (5) Differential distributions beyond angular skeleton (q^2 dependence, $B_s \rightarrow \mu\mu$ moments).

7. REPRODUCIBILITY

Capstone: lake build paper_flavor_mixing.

Synthesis witness: python3 scripts/hqiv_proof_spine_synthesis_export.py.

Attachments: data/proof_spine_synthesis_witness.json, papers/proof_spine_synthesis/scripts/docs/PROOF_SPINE_MAP.md.

SCOPE

This synthesis cites proved certificates; it does not claim PDG equality for every slot, complete SMEFT phenomenology, or closed formal QFT where the spine map marks *scaffold*.

REFERENCES

1. John Sievers, *Topological drag and stiffness in the octonionic vacuum: A geometric origin of the $SU(3)$ mass gap*, Independent research preprint, April 2026, Octonionic Hopf fibration $S^7 \hookrightarrow S^{15} \rightarrow S^8$; two-channel drag/stiffness glueball backreaction.